# CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

## Program 13 : Python Program to Implement Minimax Algorithm

### *Aim*

To develop a Python program that finds the best value of a game tree using the Minimax algorithm.

### *Algorithm*

- Define a recursive function minimax(depth, is_max) for the game tree.
- If the maximum depth (3) is reached, return the leaf value.
- At a MAX node, evaluate both child nodes and return the maximum value.
- At a MIN node, evaluate both child nodes and return the minimum value.
- Call the function from the root (depth 0) as a MAX node and display the best value.

### *Program*

```
def minimax(depth, is_max):
    if depth == 3:
        return depth

    if is_max:
        return max(minimax(depth + 1, False),
                   minimax(depth + 1, False))
    else:
        return min(minimax(depth + 1, True),
                   minimax(depth + 1, True))

print("Best value:", minimax(0, True))
```

### *Output*

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/shaik/OneDrive/Documents/13.py ===============
Best value: 3
>>>
```

### *RESULT*

Thus, the Minimax algorithm was implemented successfully using Python.